

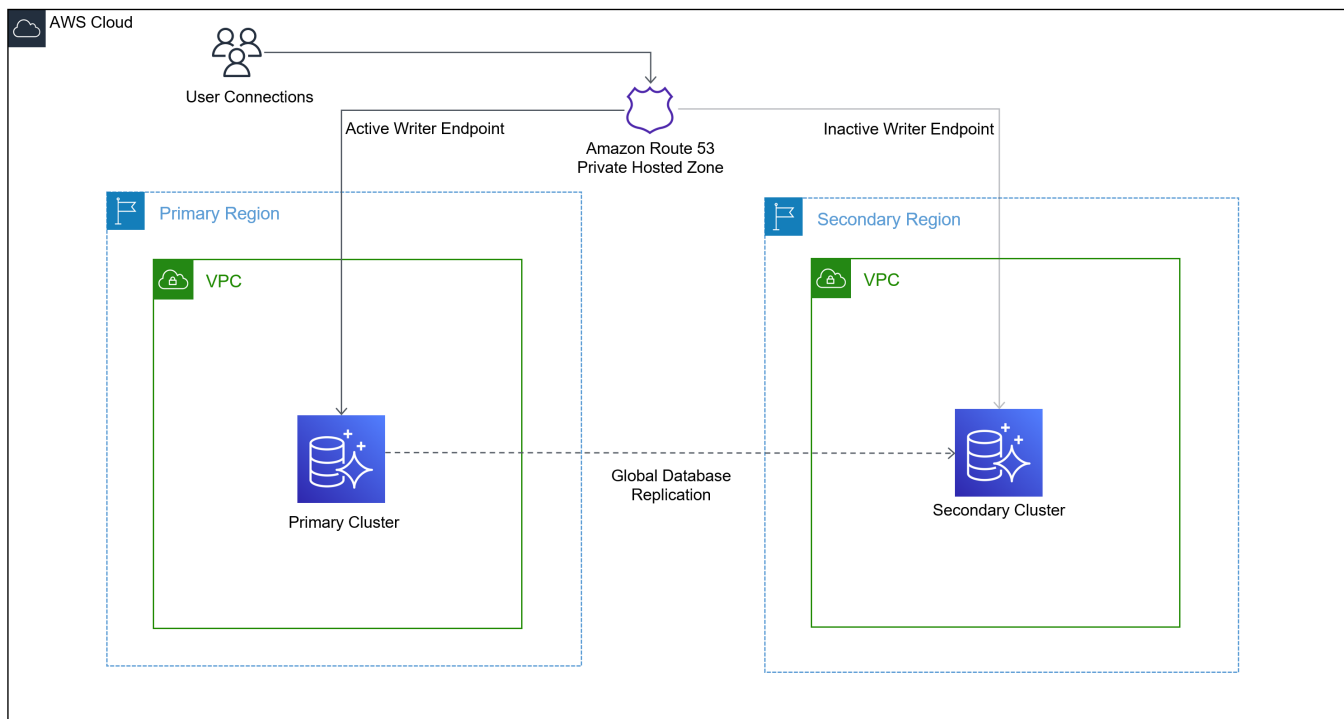
PSP - Global database endpoint for Aurora Postgres

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Overview

In the global database topology, only one Region is primary and all other Regions are secondary. The primary Region contains the only writer instance and also contains the only active writer endpoint. The writer endpoint always points to the active writer node. All secondary Regions also have writer endpoints, but they're inactive.

PSP will be using route 53 CNAME endpoint to connect to the database writer. The automation will take care of updating the CNAME to the writer endpoint of the current primary cluster in the global database in the event of a managed planned failover. There would be change required to current connection configuration and then it would be as below



Prerequisites

- AWS CLI already configured with administrator permission
- The [latest version of Python 3](#)
- The [AWS SDK for Python \(boto3\)](#)
- An AWS account with at least one Aurora global database with at least two Regions
- The [Git command line tools installed](#)

Deployment

- Create cloudformation stack

```
python3 buildstack.py --template-body 'managed-gdb-cft.yml' --stack-name 'ips-pds-global-endpoint' --consent-anonymous-data-collect 'no' --region-list 'us-east-2,us-west-2'
Started building stackid 7f667880-4b8c-11ed-99fa-02656e2bc2e4 in Region us-east-2 at: 2022-10-14T06:50:32.276253Z
Started building stackid 80a2c910-4b8c-11ed-8b98-068bb64d3291 in Region us-west-2 at: 2022-10-14T06:50:34.407379Z
Stack creation in progress for region... us-east-2
Stack creation in progress for region... us-west-2
Stack creation in progress for region... us-east-2
Stack creation in progress for region... us-west-2
Stack creation in progress for region... us-east-2
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Stack creation in progress for region... us-west-2
Stack creation in progress for region... us-east-2
Stack creation in progress for region... us-west-2
Stack creation in progress for region... us-east-2
Stack construction completed for region... us-east-2
Stack creation in progress for region... us-west-2
Stack construction completed for region... us-east-2
```

- Below CF stack will be create in us-west and us-east regions.

Stack name	Status	Created time	Description
ips-pds-global-endpoint	CREATE_COMPLETE	2022-10-14 12:20:32 UTC+0530	Creates supporting resources for automated global database endpoint management solution.

- Create global endpoint for Aurora Postgres Global database

```
python3 create_managed_endpoint.py --cluster-cname-pair '{"psp-test-global":"psp-test-db.psp-ppd.a.intuit.com"}' --hosted-zone-name=psp-ppd.a.intuit.com --region-list 'us-west-2,us-east-2'

Processing region us-west-2 :
Processing regional cluster psp-test :
Hosted Zone psp-ppd.a.intuit.com. already exists. Checking vpcs..
VPC vpc-02eb1ccd0822b6f9e Already exists. checking if CNAME exists..
CNAME psp-test-db.psp-ppd.a.intuit.com doesn't exist. Adding CNAME record..
Cname psp-test-db.psp-ppd.a.intuit.com Successfully created with endpoint psp-test.cluster-ccqjgvvo0rwy.us-west-2.rds.amazonaws.com
Added entry to the dynamodb table for cluster psp-test
Processing regional cluster psp-testdr :
No entries made for this cluster. Cluster is not in the current region.

Processing region us-east-2 :
Processing regional cluster psp-test :
No entries made for this cluster. Cluster is not in the current region.
Processing regional cluster psp-testdr :
Hosted Zone psp-ppd.a.intuit.com. already exists. Checking vpcs..
VPC vpc-03af2ef666f86d6d2 Already exists for secondary region cluster psp-testdr
Added entry to the dynamodb table for cluster psp-testdr
```

- Below will be created and available to use
 - Global resources created per account:
 - Private hosted zone (Route 53)** – A private hosted zone is created based on the values you passed.
 - CNAME records** – A CNAME record is created inside the hosted zone based on the parameters you passed per cluster.
 - Local resources created per Region:
 - IAM role** – An [AWS Identity and Access Management](#) role is created so the Lambda function can assume this role while running. The role is very granular and minimalistic. It selectively grants only permissions needed to carry out the functions needed to run the Lambda function.
 - Lambda function** – This function is invoked on a global database failover completion event, and updates the CNAME. It contains the core logic of the solution automation.
 - DynamoDB table** – A DynamoDB table named gdbcnamepair is created. This table keeps track of the clusters that are managed by this solution.
 - EventBridge rule** – This rule is triggered when a global database completes failover in the Region. This rule has the Lambda function as its target.

The screenshot displays two AWS Management Console pages. The top page shows the 'Hosted zones' for 'psp-ppd.a.intuit.com'. It lists four records:

Record name	Type	Routing policy	Value/Route traffic to	TTL	Health	Evaluation
psp-ppd.a.intuit.com	NS	Simple	ns-1536.awsdns-00.co.uk, ns-0.awsdns-00.com, ns-1024.awsdns-00.org, ns-512.awsdns-00.net	172800	-	-
psp-ppd.a.intuit.com	SOA	Simple	ns-1536.awsdns-00.co.uk. awsdns-hostmaster.amazon.com. 1 7200 900 1209600 ...	900	-	-
ppsp-pds-db.psp-ppd.a.intuit.com	CNAME	Simple	ppsp-pds-db.cluster-ccqjgvvo0rwy.us-west-2.rds.amazonaws.com	1	-	-
psp-test-db.psp-ppd.a.intuit.com	CNAME	Simple	psp-test.cluster-ccqjgvvo0rwy.us-west-2.rds.amazonaws.com	1	-	-

The bottom page shows the 'gdb-managed-endpoint' Lambda function overview. It includes a diagram showing the function triggered by 'EventBridge (CloudWatch Events)'. The description states: 'Lambda function that manages the writer endpoint, when the global database failover completes in this region. This Lambda is triggered by an event bridge rule.' The function ARN is 'arn:aws:lambda:us-east-2:152430470825:function:gdb-managed-endpoint'.

- Whitelist newly created private Hosted Zone for all your IKS services(upstream and downstream etc) in all regions**, you need to work with MSaaS team on this by following steps as mentioned below
 - Kindly create a JIRA referring this [Private Hosted Zone Whitelisting for IKS](#)
 - Request your application architect to provide go ahead and approval in the JIRA for the concerned whitelisting
 - Complete till Step 4 by following [How to associate a Route 53 private hosted zone with a VPC on a different AWS account](#) document
 - Kindly note that you use your IKS VPC IDs in the command for all applicable AWS regions
 - Start a thread by pinging in **#msaas-support** Slack channel
 - Ensure that AWS account VPCs(in all applicable regions) are allowed to talk to respective IKS regions VPCs through ITH/TGW**
 - Here is a [sample JIRA](#) for enabling VPC traffic over TGW/ITH
Sample [Sample JIRA2](#)
 - For any AWS related issue or query, please start a thread in **#team-aws-help** Slack channel
- Try to login to your PG Aurora DB using the newly created CNAME

Switchover

CNAME record before switchover

☐	Record name	Type	Routin...	Differ...	Alias	Value/Route traffic to	TTL (s...	Health ...	Evalu...	Record ID
☐	psp-ppd.a.intuit.com	NS	Simple	-	No	ns-1536.awsdns-00.co.uk. ns-0.awsdns-00.com. ns-1024.awsdns-00.org. ns-512.awsdns-00.net.	172800	-	-	-
☐	psp-ppd.a.intuit.com	SOA	Simple	-	No	ns-1536.awsdns-00.co.uk. awsdns-hostmaster.amazon.com. 1 7200 900 1209...	900	-	-	-
☐	ppsp-pds-db.psp-ppd.a.intuit.com	CNAME	Simple	-	No	ppsp-pds-db.cluster-ccqjgvvo0rwy.us-west-2.rds.amazonaws.com	1	-	-	-
☐	psp-test-db.psp-ppd.a.intuit.com	CNAME	Simple	-	No	psp-test.cluster-ccqjgvvo0rwy.us-west-2.rds.amazonaws.com	1	-	-	-

CNAME record updated after switchover via global endpoint automation

☐	Record name	Type	Routin...	Differ...	Alias	Value/Route traffic to	TTL (s...	Health ...	Evalu...	Record ID
☐	psp-ppd.a.intuit.com	NS	Simple	-	No	ns-1536.awsdns-00.co.uk. ns-0.awsdns-00.com. ns-1024.awsdns-00.org. ns-512.awsdns-00.net.	172800	-	-	-
☐	psp-ppd.a.intuit.com	SOA	Simple	-	No	ns-1536.awsdns-00.co.uk. awsdns-hostmaster.amazon.com. 1 7200 900 1209...	900	-	-	-
☐	ppsp-pds-db.psp-ppd.a.intuit.com	CNAME	Simple	-	No	ppsp-pds-db.cluster-ccqjgvvo0rwy.us-west-2.rds.amazonaws.com	1	-	-	-
☐	psp-test-db.psp-ppd.a.intuit.com	CNAME	Simple	-	No	psp-testdr.cluster-cxph5mzesrt.us-east-2.rds.amazonaws.com	1	-	-	-

Reference

- [automate-amazon-aurora-global-database-endpoint-management](#)